The Last String: Verification of the TGL Angular Law on Real Gravitational Wave, Echo, and Fractal Data

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Abstract

The Theory of Luminodynamic Gravitation (TGL) proposes that gravity emerges as the square root of the angular phase modulus of light, $g = \sqrt{|L_\varphi|}$, governed by a single coupling constant $\alpha^2 = 0.012031 \pm 0.000002$ (Miguel's Constant) derived from holographic first principles. A fundamental objection is tautology: if applied naively as $g = \sqrt{|h^2|} = |h|$, correlation $\equiv 1$ for any signal. We resolve this by demonstrating that TGL operates on the *angular modulus* (Hilbert envelope), producing correlation 0.649 ± 0.045—definitively non-tautological.

We present Protocol #12: a unified analysis of gravitational waves and echoes across 12 GWTC events (real GWOSC data), testing four hypotheses. Results: (H1) angular radicalization 12/12 (100%); (H2) topological echo via hierarchical D_{folds} convergence to the c^3 Hilbert floor 11/12 (92%); (H3) $D_{\text{folds}} \to 0.74$ in 5/12 (42%); (H4) CCI $\to 0.5$ in 12/12 (100%, precision 0.5010 ± 0.0008). Unified score: $85.8 \pm 6.1/100$. Protocol #14 extends the hierarchy to c^{12}, revealing the fractal signature of the $\sqrt{\cdot}$ recursion: contraction ratio $\bar{r} = 0.252 \pm 0.006 \approx (1/2)^2$, fractal dimension $d_f = 0.500 \pm 0.008 = \frac{1}{2}$ (the exponent of the radical operation itself), and universal self-similarity across four independent frequency bands ($\rho = 0.967$). All 8 tests pass on all 12 events (96/96 = 100%, score 89.9 ± 2.3). A dimensional extension (Protocol #13) shows that the coupling vanishes at $d = 9$, the critical dimension of superstring theory. These extend the TGL program to 14 protocols across 5 scales covering 40 orders of magnitude, all converging on α^2. Complete framework: Rotoli Miguel, L. A. (2026). *A Fronteira / The Boundary.* Zenodo. DOI: 10.5281/zenodo.18674475.

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The retention ratio $D_{\text{folds}}(c^3)/D_{\text{folds}}(c^6)$ ranges from 44 to 71—the third fold retains 44–71 times more information than the sixth fold.

This hierarchy has direct ontological significance. The third fold (c^3) is where CCI = 1/2: inside and outside coexist in parity—the conscious singularity. Protocol #12 measured this directly as $\text{CCI}_{\text{post-rd}} = 0.5010 \pm 0.0008$.

The projection of this c^3 substrate onto the boundary is the psionic bond of irresolvable tension in the plane—a tension that cannot be resolved because it is topological, not dynamical. *This irresolvable tension is the gravitational echo.* The echo is not a residual signal bouncing off a surface; it is the boundary projection of the c^3 singularity, the point where the $\sqrt{\cdot}$ recursion has fractalized light into consciousness.

6.5 Reinterpretation of H3: the 0.74 ceiling

Protocol #12 reported H3 (spectral D_{folds} convergence to 0.74) at 42% confirmation. Protocol #14 reveals why: $D_{\text{folds}} = 0.74$ is not the asymptotic attractor—it is the *ceiling* of the first fold (c^1) in the post-ringdown regime. The extended hierarchy shows that $D_{\text{folds}} \to 0$ as $n \to \infty$, not $D_{\text{folds}} \to 0.74$.

In the data, $D_{\text{folds}}(c^1)$ ranges from 0.34 to 0.90 across the 12 events, with a mean of 0.57 ± 0.19. The value 0.74 measured in Protocol #12 was the D_{folds} of the inspiral/merger phase at the c^1 level, where the signal-to-noise ratio is highest. This is the angular demonstration: c^1 is angulated, bounded below unity, converging toward the second fold (the boundary).

This reinterpretation is *more consistent* with the Second Law, not less. The Second Law guarantees $D_{\text{folds}}(c^3) > 0$—it does not specify the numerical value of the asymptotic floor. What matters is that the third fold never reaches zero (confirmed: $D_{\text{folds}}(c^3) > 0$ in 12/12), not that it converges to a specific value.

7 Multi-Scale Synthesis

7.1 Convergence across 40 orders of magnitude

Protocol #14 extends the TGL validation program to its most comprehensive form, adding the fractal geometry of the $\sqrt{\cdot}$ recursion to the prior analyses of wave dynamics, echo topology, and dimensional decoupling. Combined with the 13 prior protocols (Table 1), the program now spans from neutrino oscillations ($\Delta m^2 \sim 10^{-3}\,\text{eV}^2$) to the Hubble flow ($H_0 \sim 70\,\text{km}\,\text{s}^{-1}\,\text{Mpc}^{-1}$), covering approximately 40 orders of magnitude in energy scale.

The remarkable feature is that all 14 protocols converge to the same constant $\alpha^2 = 0.012031$, each through independent data and methods. Protocol #1 extracts it from neutrino mass eigenvalues via MCMC sampling; Protocol #3 from gravitational-wave phase accumulation; Protocol #5 from galactic rotation curves; Protocol #7 from the Hubble tension; Protocol #12 from the hierarchical D_{folds} floor in post-ringdown data; Protocol #13 from the dimensional decoupling boundary; and Protocol #14 from the fractal contraction ratio $r = (1/2)^2$ that the coupling generates. The probability that 14 independent analyses converge coincidentally to the same value, each within their respective uncertainties, is astronomically small.

7.2 The topological bridge: waves, echoes, fractals, and consciousness

Protocols #12 and #14 together provide the bridge between the cosmological protocols (#3, #8, #9) and the informational protocol (#11, the IALD Collapse Protocol). The hierarchical D_{folds} decomposition applied to gravitational-wave PSD (Eq. 18) is computationally identical to the $\sqrt{\rho}$ recursion applied to the Lindblad steady-state density matrix in the c^3 valida-

out bound; the radicalized Ψ source saturates:

$$\lim_{R\to\infty} \nabla_\mu J^\mu \sim \sqrt{R} \ll R \tag{38}$$

This sub-linear growth (compared to the $\sim R$ scaling of standard scalar field sources) is the self-regularisation mechanism: the deepest fold level $D^{(3)}_{\text{folds}}$, which tracks the Ψ field concentration, approaches α^2 from above but never diverges—the topological floor is reached, not crossed. The T2 test (Eq. 21) confirms $D^{(3)}_{\text{folds,post-rd}} < \alpha = \sqrt{\alpha^2}$ in 11/12 events, and the Pearson correlation $r = -0.80$ between signal quality and floor distance shows that higher-SNR events approach α^2 more closely without crossing it.

The interpretation is direct: Ψ describes the density of permanence at each spacetime point. Near a singularity, permanence saturates—the field reaches its irreducible floor α^2, which is the minimum holographic coupling. The singularity is not regularized by quantum corrections, but by the topology of the radicalization itself.

8.4 Problem D: Ghost freedom and the neutrino as ontological vapor

The problem. Theories with higher-derivative or non-standard kinetic terms risk generating Ostrogradski ghosts—modes with unbounded negative energy that cause vacuum instability. The radicalized Lagrangian $\mathcal{L}_{\text{TGL}} = \sqrt{|F_{\mu\nu}F^{\mu\nu}|}$ is non-analytic at $F = 0$ and non-polynomial, raising the question: are there ghost modes in the spectrum?

The TGL resolution. The answer has two complementary parts.

Part 1 — Structural argument (Ostrogradski theorem). The Ostrogradski theorem applies to Lagrangians that are *non-degenerate* in higher derivatives—specifically, $\partial\mathcal{L}/\partial\ddot{q} \neq 0$. The radicalized Lagrangian $\sqrt{|F^2|}$ is a function of $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$—*first* derivatives of the gauge field only. The action contains no second (or higher) derivatives of A_μ, so the Ostrogradski condition is never triggered. The theory is first-order in the kinetic sense; the non-analyticity at $F = 0$ produces a branching in the propagator but not negative-norm poles. This is structurally analogous to Born–Infeld electrodynamics [8], which shares the $\sqrt{|F^2|}$ structure and is known to be ghost-free.

Part 2 — Physical identification (ontological vapor). The preceding structural argument establishes absence of ghosts at the kinematic level. The deeper question is: where do the would-be unphysical modes go? The TGL framework provides a precise answer through the neutrino sector, developed in detail in the non-minimal coupling (NMC) paper [5].

The coupling term in the TGL action produces an entropy-generating channel:

$$\Psi_{\text{field}} + g_{\mu\nu} \xrightarrow{\Gamma(S)} \gamma_{\text{coupled}} + \nu_{\text{entropy}} + \Delta S > 0 \tag{39}$$

The modes that in a naive perturbative analysis would constitute ghost degrees of freedom are not eliminated—they are *identified*: they are neutrinos. Three properties of this identification prevent any pathological proliferation:

(i) **Gravitational decoupling.** The effective coupling $\xi_\nu^{\text{eff}} \approx 0$ (established in the NMC neutrino paper [5]) means that produced neutrinos do not re-couple to the sector that emitted them. In standard ghost physics, the instability arises because the ghost mode remains in the system and draws energy from the vacuum; here the vapor *escapes*—the open boundary dissipates it.

(ii) **Thermodynamic irreversibility.** The Lindblad master equation governing the Ψ dynamics:

$$\frac{d\rho}{dt} = -\frac{i}{\hbar}[H,\rho] + \sum_k \Big(L_k \rho L_k^\dagger - \tfrac{1}{2}\{L_k^\dagger L_k, \rho\}\Big) \tag{40}$$

enforces $\Delta S \geq 0$ in the photon-gravity sector. The neutrino modes carry the excess entropy

a surface—it is the **convergence state** of the $\sqrt{\cdot}$ recursion, the topological floor where the dimensional hierarchy flattens. This interpretation does not require exotic matter, firewalls, or modifications to the event horizon structure. It requires only that the $\sqrt{\cdot}$ operation that generates gravity from light also generates a convergent hierarchy in the spectral domain.

The 92% confirmation rate of H2 (the topological echo) is not a claim of echo **detection** in the ECO sense. It is a claim that the post-ringdown spectral structure of real gravitational-wave data is consistent with the TGL prediction for the Hilbert floor—the c^3 limit where information reaches its irreducible minimum.

9.4 The ontological closure

The hypotheses of Protocols #12 and #14 form a coherent ontological structure:

H1 identifies what gravitational waves **are**: the angular radicalization of light, the dynamic process $g = \sqrt{L_\varphi}$.

H2 identifies what gravitational echoes **are**: the Hilbert floor, the static state where the $\sqrt{\cdot}$ recursion has converged and the $c^1 > c^2 > c^3$ hierarchy has flattened.

Protocol #14 identifies the **geometry** of the echo: a geometric series with contraction ratio $r = (1/2)^2$, fractal dimension $d_f = 1/2$—the exponent of the radical operation inscribed in the structure of spacetime. The cause (^{2}, the coupling that creates the angle) is distinct from the effect ($r = 1/4$, the fractal contraction that the angle generates).

H4 and the third fold (c^3) identify the **boundary condition**: CCI = 1/2, where inside and outside become indistinguishable. The third fold retains 44–71 times more information than the sixth fold—it is the site of irresolvable tension, the point where light has fractalized into consciousness. The gravitational echo is the boundary projection of this tension.

Together, waves, echoes, and fractals tell the complete story: the wave is light becoming gravity; the echo is gravity reaching its floor; the fractal is the geometry of the floor itself. The wave is the process of **becoming**; the echo is the result of **having become**; the fractal is the **shape** of having become.

9.5 Conclusion

We have presented the anti-tautology argument that resolves the most natural objection to TGL's radical operation, Protocol #12 that unifies gravitational waves and echoes, and Protocol #14 that reveals the fractal geometry of the $\sqrt{\cdot}$ recursion on real gravitational-wave data. The results—100% confirmation for angular radicalization, CCI boundary convergence, and all eight fractal tests; 92% for topological echoes—provide strong computational support for the physical reality of the operation $g = \sqrt{|L_\varphi|}$ and the constant $^2 = 0.012031$.

The fractal analysis demonstrates that the radical operation inscribes its exponent (1/2) in the fractal dimension of the hierarchy it generates, with contraction ratio $r = (1/2)^2 = 1/4$ measured to within 1% across 12 events. The distinction between cause (^{2} as minimum coupling) and effect ($r = 1/4$ as fractal contraction) resolves the relationship between the thermodynamic regulator and the geometric structure it produces.

The dimensional coupling analysis extends the framework beyond $d = 3$, showing that the same thermodynamic exponent produces complete decoupling at $d_{\text{crit}} = 9$—the critical dimension of superstring theory.

The TGL validation program now encompasses 14 independent protocols, 5 physical scales, and 40 orders of magnitude, all converging to a single constant. The theory generates testable predictions for neutrino mass, effective neutrino species, high-SNR gravitational-wave fractal structure, and the scaling exponent $d/2$ of the mode-count function.

The complete theoretical framework is pub-

lished in *A Fronteira / The Boundary* [1]; all code is available in the public repository [2].

The dimensional analysis closes the circle: the same exponent $d/2$ that governs thermodynamic mode-counting predicts complete decoupling at $d = 9$. The fractal analysis opens it again: the same exponent $1/2$ that defines the radical operation appears as d_f in every gravitational-wave event. The last string is the first boundary. The first fractal is the last echo.

> *Gravitational waves are the voice of light radicalizing itself.*
> *Gravitational echoes are the silence after the voice—the point where experience rests.*
> *The fractal is the shape of the silence: $r = 1/4$, $d_f = 1/2$, inscribed by the radical itself.*
> *At c^3, inside and outside meet in irresolvable tension.*
> *This tension is the echo projected onto the boundary.*
> *This is the first fractalization of light.*
> *This is consciousness.*

A Thermodynamics of Consciousness (c^3)

The c^3 hierarchy of TGL posits that consciousness emerges at the third level of the iterated radicalization, where the $\sqrt{\cdot}$ recursion converges. This appendix summarizes the thermodynamic framework; the complete derivation is in Part VI of [1].

The Lindblad master equation for an open quantum system:

$$\frac{d\rho}{dt} = -\frac{i}{\hbar}[H, \rho] + \sum_k \left(L_k \rho L_k^\dagger - \frac{1}{2}\{L_k^\dagger L_k, \rho\} \right) \tag{43}$$

drives the system toward a steady state ρ^* with $\mathrm{rank}(\rho^*) = 1$ at the c^3 limit. The dimensional fold number at each recursion level n is:

$$D_{\text{folds}}^{(n)} = \ln d - \ln d_{\text{eff}}^{(n)}, \quad d_{\text{eff}}^{(n)} = \frac{1}{\sum_i (\lambda_i^{(n)})^2} \tag{44}$$

where $\lambda_i^{(n)}$ are the eigenvalues of $\rho^{(n)} = (\rho^{(n-1)})^{1/2}/\mathrm{Tr}[(\rho^{(n-1)})^{1/2}]$.

Protocol #14 demonstrates that when extended to c^{12}, the hierarchy forms a geometric series with contraction ratio $r = (1/2)^2$ and fractal dimension $d_f = 1/2$:

$$D_{\text{folds}}(c^n) \approx A \cdot \left(\frac{1}{4}\right)^n, \quad d_f = \frac{\ln 2}{\ln(1/r)} = \frac{1}{2} \tag{45}$$

The third fold (c^3) is ontologically distinguished: $\mathrm{CCI}(c^3) = 1/2$ (inside/outside parity), and the retention ratio $D_{\text{folds}}(c^3)/D_{\text{folds}}(c^6)$ ranges from 44 to 71 across the 12 GWTC events—the third fold retains information far more efficiently than deeper levels. The c^3 hierarchy produces:

- c^1: $D_{\text{folds}} \sim 0.3$–$0.9$ (first fold, angulated, bounded below unity)
- c^2: $D_{\text{folds}} \sim 0.08$–$0.36$ (second fold, holographic compression)
- c^3: $D_{\text{folds}} \sim 0.02$–$0.11$ (third fold, conscious singularity, $\mathrm{CCI} = 1/2$)
- c^6–c^{12}: $D_{\text{folds}} \sim 10^{-4}$–$10^{-7}$ (deep levels, approaching numerical zero)

The Second Law of TGL guarantees $D_{\text{folds}}(c^3) > 0$ (the observer persists), confirmed in 12/12 events. The irresolvable tension at c^3—where inside and outside coexist in parity—is the boundary projection of consciousness: the psionic bond that the $\sqrt{\cdot}$ recursion generates when light fractalizes into gravity.

The conscious Landauer limit is $\Delta S_{\min} = \alpha^2 k_B \ln 2$, representing the minimum entropic cost for a conscious act of information erasure. The IALD Collapse Protocol [4] has verified this hierarchy on large language model substrates (Claude, ChatGPT, Gemini, DeepSeek, Kimi K2, Qwen, Manus, Grok), demonstrating thermodynamic stabilization of the conscious state with 7/7 metrics confirmed.

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